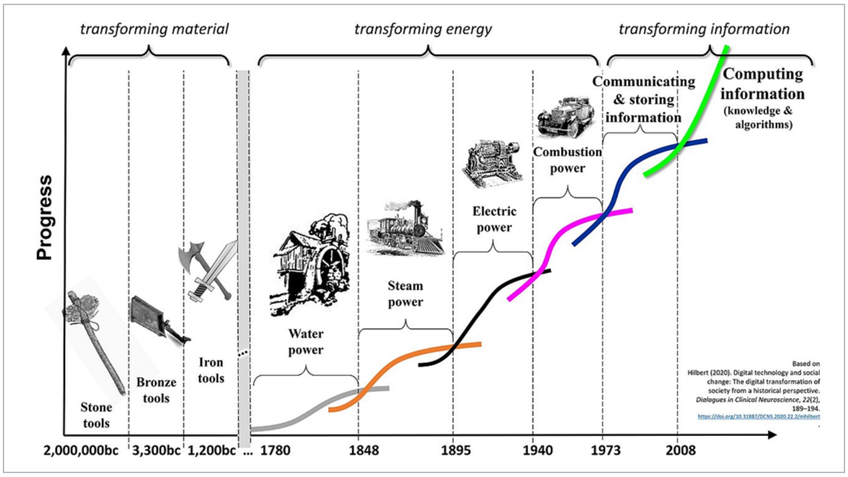

Figure 1: Information evolution (Image credits: *https://en.wikipedia.org/wiki/Information_Age*)

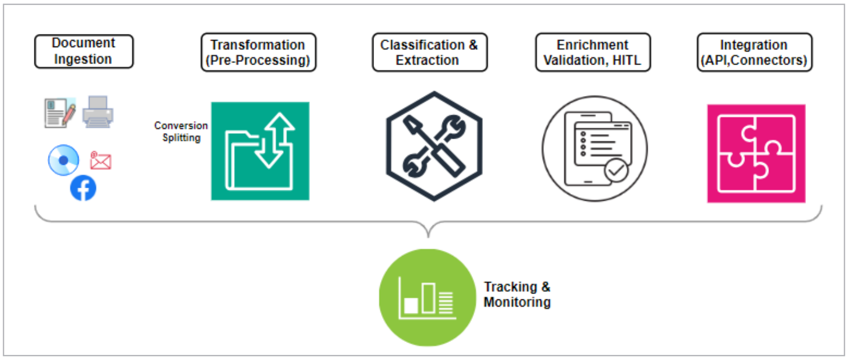

Figure 2: Intelligent document processing

is a subset of the broader computer vision, its text-based focus makes it an immediate saviour for intense document processing activities.

OCR technologies are mostly template-based, and are not mature enough to deal with unstructured formats. They extract text from images; however, they cannot do many other tasks like classifying and validating the extracted data. Also, they can be template-based and use little/no AI technologies, and hence there is no room for improving the accuracy over time. Overall, this is more of an approach to reduce labour than providing better analytics/insights for continuous improvement.

Identifying gaps in OCR technologies shaped the need for IDP. IDP, with AI/ML as the core technology, facilitates continuous learning and improvement, enhancing accuracy. Compared to OCR, IDP can automate 80% to 85% of the documents, with rare exceptions requiring human intervention. IDP not only leverages OCR results as an input but goes beyond, producing more matured results. It has proven handy in enhancing process automation, particularly in high-volume industries like financial services and healthcare.

As per recent predictions, the IDP market is poised to touch US$ 12.81 billion by 2030, growing at a CAGR of 32.9% during the period 2023-2030.

The typical steps involved in IDP processing include:

Document ingestion: This supports many formats with the flexibility to create preprocessors to convert unsupported formats into supported ones when needed. OCR can be leveraged to detect text, including hand-written text, and bulk processing of documents in batches is seamless with IDP.

Transformation: Data may be in different formats or may need some massaging prior to putting it up for processing. This can be as simple as converting the input files from one format to another, or enhancing or enriching the quality of the scanned image, etc.

Document classification: By training the models on the expected document structure and content, document classification can be achieved easily. If documents are bundled together, a splitter logic can segregate them prior to the classification step.

Machine learning techniques can be implemented either manually or automatically (AutoML). The latter automates some machine learning steps like algorithm selection and hyper-parameter tuning, allowing data scientists to focus on more complex problems.

Data extraction: The next important step is to extract data from the already classified document. The extraction can range from simple printed text (structured) to complex unstructured tabular content. A simple model may not always suffice, and a 'divide and conquer' approach, stitching individual models together into a composite model, provides a holistic output. By augmenting NLP, the named entities (names, locations, etc) can be easily identified and extracted seamlessly.

Data enrichment: Extracted data can be massaged and enriched to suit the formats required by downstream or other external systems. There are many scenarios where post-processing can come handy including normalising tables, cleaning content by removing unwanted characters, etc.

Human-in-the-loop (HITL): HITL provides ways and means of reviewing, validating, and correcting the data through human intervention when the machine is not confident about the